226868 and other stars in its vicinity, measurements crucial to avoiding severe pitfalls in estimating the mass of Cyg X-1.

The theoretical astrophysicists contributing to the effort included people like Zel'dovich and Novikov, who proposed the method of search; Bohdan Paczynski, Yoram Avni, and John Bahcall, who developed complex but reliable ways to circumvent the mass-estimate pitfalls; Geoffrey Burbidge and Kevin Prendergast, who realized that the hot, X-ray-emitting gas should form a disk around the hole; and Nikolai Shakura, Rashid Sunyaev, James Pringle, Martin Rees, Jerry Ostriker, and many others, who developed detailed theoretical models of the X-ray-emitting gas and its disk, for comparison with the X-ray observations.

By 1974 this massive effort had led, with roughly 80 percent confidence, to the picture of Cyg X-1 and its companion star HDE 226868 that is shown in an artist's sketch in the right half of Figure 8.7. It was just the kind of picture that Zel'dovich and Novikov had envisioned, but with far greater detail: The black hole at the center of Cyg X-1 has a mass definitely greater than 3 Suns, probably greater than 7 Suns, and most likely about 16; its optically bright but X-ray-dark companion HDE 226868 has a mass probably greater than 20 Suns and most likely about 33, and it is roughly 20 times larger in radius than the Sun; the distance from the star's surface to the hole is about 20 solar radii (14 million kilometers); and the binary is about 6000 light-years from Earth. Cyg X-l is the second brightest object in the X-ray sky; HDE 226868, while very bright in comparison with most stars seen by a large telescope, is nevertheless too dim to be seen by the naked eye.

In the two decades since 1974, our confidence in this picture of Cyg X-l has increased from roughly 80 percent to, say, 95 percent. (These are my personal estimates.) Our confidence is not 100 percent because, despite enormous efforts, no unequivocal signature of a black hole has yet been found in Cyg X-l. No signal, in X-rays or light, cries out at astronomers saying unmistakably, "I come from a black hole." It is still possible to devise other, non-black-hole explanations for all the observations, though those explanations are so contorted that few astronomers take them seriously.

By contrast, some neutron stars, called pulsars, produce an unequivocal "I am a neutron star" cry: Their X-rays, or in some cases radio waves, come in sharp pulses that are very precisely timed. The timing is as precise, in